MILITARY SPECIFICATION SHEET

FILTERS AND CAPACITORS, RADIO FREQUENCY/ELECTROMAGNETIC INTERFERENCE SUPPRESSION, HERMETICALLY SEALED, STYLE FS66

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the filters described herein shall consist of this specification and the latest issue of MIL-F-28861.

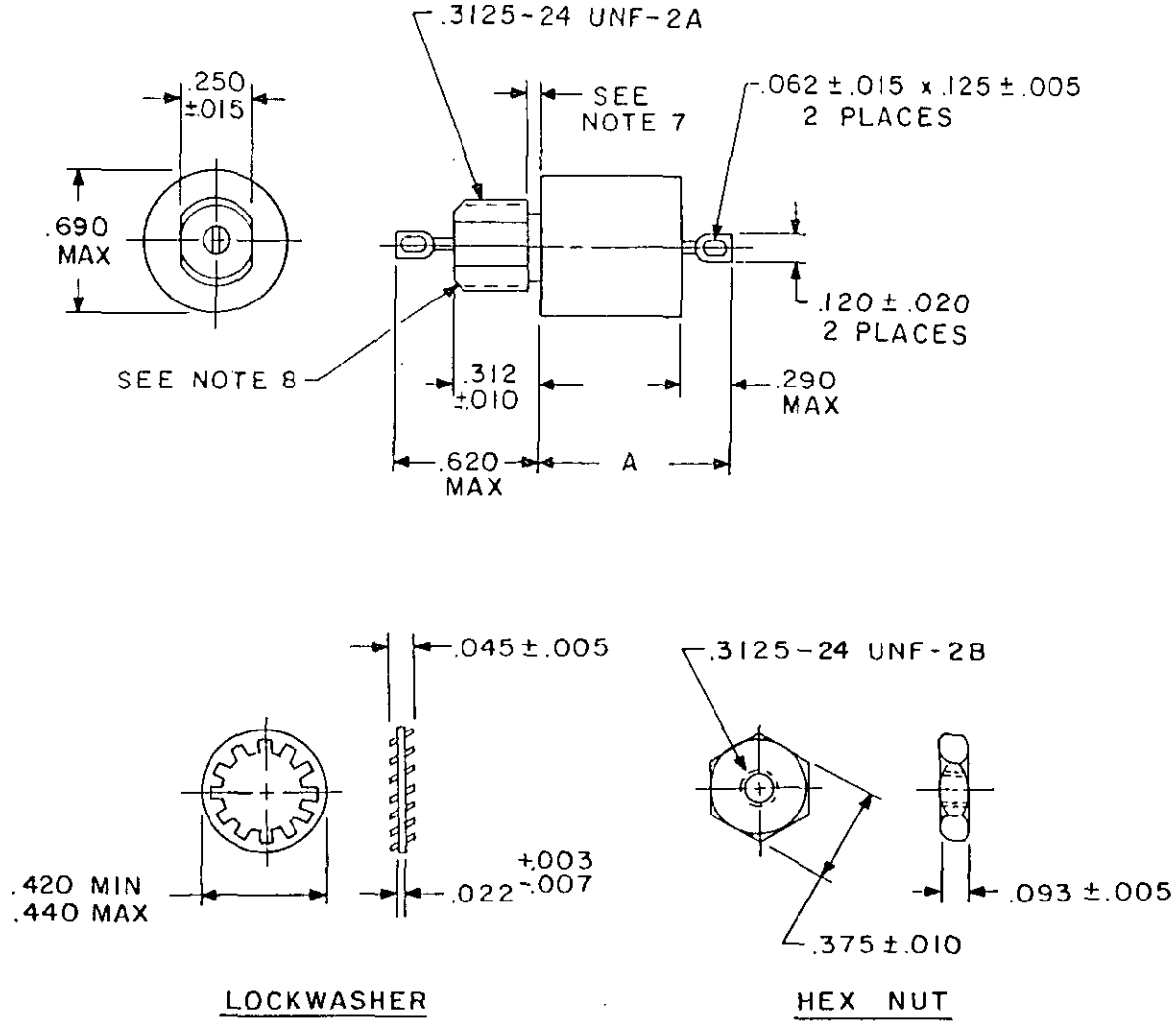
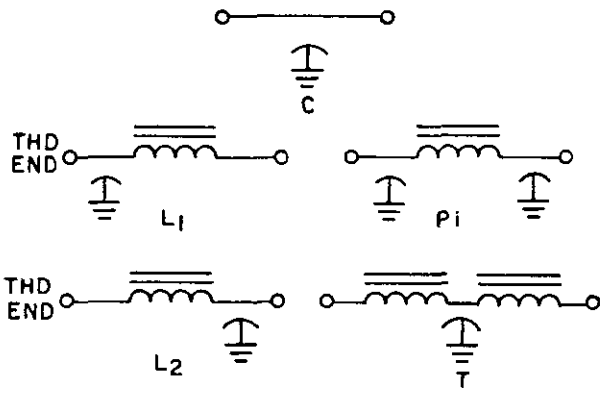


FIGURE 1. Case and hardware dimensions and circuit diagrams.

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CIRCUIT DIAGRAMS

Dash no.	A dimension max	Weight max (grams)
001	.700	19
002 thru 011	1.060	21
012 thru 016	1.205	30
017 thru 019	1.300	30

Inches	mm	Inches	mm
.003	0.08	.250	6.35
.005	0.13	.290	7.37
.007	0.18	.312	7.92
.010	0.25	.375	9.53
.015	0.38	.420	10.67
.020	0.51	.440	11.18
.022	0.56	.620	15.75
.045	1.14	.690	17.53
.062	1.57	.700	17.78
.093	2.36	1.060	26.92
.120	3.05	1.205	30.61
.125	3.18	1.300	33.02

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Circuit diagram is for information only.
4. All filters shall be supplied with mounting hardware (hex nut and lockwasher). Mounting hardware shall be furnished with the same finish as the filter case.
5. Recommended mounting torque: 60 oz-in. ±4 oz-in.
6. Terminal identification (nonsymmetrical filters): The case shall be marked at threaded end of the filter, with the symbol "C" or the symbol "L", as follows:

Circuit	Symbol
L_1	C
L_2	L

7. Imperfect thread or undercut optional .062 inch (1.57 mm) maximum.
8. One imperfect thread allowed .035 inch (0.89 mm) maximum.

FIGURE 1. Case and hardware dimensions and circuit diagrams - Continued.

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REQUIREMENTS:

Design and construction:

Dimensions and configuration: *See figure 1.*

Case finish: T only (tin-plated or tin-lead plated).

Terminals: *Solderable.*

Operating temperature range: -55°C to +125°C.

Rated voltage: 400 V dc/230 V ac, 0 to 400 Hz.

Rated current: See table I.

Capacitance to ground: See table I for capacitance value.

Dissipation factor: 3.0 percent maximum.

Voltage and temperature limits of capacitance: +15, -40 percent.

Insulation resistance:

At +25°C: 1,000 megohm, minimum.

At +125°C: 100 megohms, minimum.

Insertion loss: See table I.

Voltage drop: See table I, measured at 400 Hz.

DC resistance: See table I.

Seal: In accordance with MIL-F-28661.

Temperature rise: +25°C, maximum.

Solderability of terminals: In accordance with MIL-F-28861.

Product assurance level: Class B only.

Part number: *The part number shall be as follows:*

M28861/16- XXX T B

XXX: Dash no. per figure 1 and table I

T: Case finish

B: Product assurance level per MIL-F-28861

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TABLE I. Electrical characteristics.

Dash number	Circuit	Maximum rated current (amps)	Capacitance -0, +100 (μF) percent	Maximum voltage drop (volts) (ac rms)	Maximum voltage drop (volts) (dc)	Maximum dc resistance (ohms)	Minimum insertion loss (dB) in accordance with MIL-STD-220 1/ 2/ At +25°C 100 kHz	150 kHz	300 kHz	1 MHz	10 MHz	100 MHz	1 GHz	At -55°C and +125°C 100 kHz	150 kHz	300 kHz	1 MHz	10 MHz	100 MHz	1 GHz
001	C	15	.15	.14	.12	.008	7	10	16	25	40	52	70	5	8	14	24	38	50	70
002	L1																			
003	L2	.5	.15	.47	.165	.33	18	24	35	56	60	70	70	16	23	35	56	70	70	70
004	L1																			
005	L2	1.0	.15	.47	.15	.15	12	19	31	52	60	70	70	10	17	29	50	70	70	70
006	L1																			
007	L2	3.0	.15	.32	.078	.026	7	11	19	36	70	70	70	5	9	17	34	70	70	70
008	L1																			
009	L2	5.0	.15	.21	.065	.013	7	10	16	28	64	70	70	5	8	14	26	62	70	70
010	L1																			
011	L2	10.0	.15	.12	.08	.008	7	10	16	25	53	70	70	5	8	14	23	51	70	70
012	Pi	.5	.20	.47	.165	.33	24	34	52	80	80	80	80	22	32	50	80	80	80	80
013	Pi	1.0	.20	.47	.15	.15	16	27	46	77	80	80	80	14	25	44	75	80	80	80
014	Pi	3.0	.20	.32	.078	.026	-	-	31	63	80	80	80	-	-	29	61	80	80	80
015	Pi	5.0	.20	.21	.065	.013	-	-	12	52	80	80	80	-	-	10	50	80	80	80
016	Pi	10.0	.20	.12	.08	.008	-	-	-	31	80	80	80	-	-	-	38	80	80	80
017	T	1.0	.15	.21	.052	.052	6	12	25	54	70	70	70	4	10	23	52	70	70	70
018	T	2.0	.15	.22	.076	.038	6	10	18	41	70	70	70	4	8	16	39	70	70	70
019	T	4.0	.15	.32	.104	.026	6	10	16	31	70	70	70	4	8	14	29	70	70	70

1/ Insertion loss measurements shall be made under full-load over the frequency range of 100 kHz to 10 MHz. Insertion loss measurements above this frequency range shall be under no-load.

2/ The insertion loss requirements between any two adjacent specified frequencies shall be that of the lower of the two frequencies in order to accommodate resonant dips.

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Cataloging information: Dash number 001 shall be cataloged under FSC 5910 as a feed-through ceramic capacitor. Dash numbers 002 through 019 shall be cataloged under 5915 as radio frequency interference filters.

SUPERSESSION: This specification sheet supersedes drawing number 84084, FSCM 14933 when a qualified source is available.

Custodians:
Army - ER
Navy - EC
Air Force - 85
NASA - NA

Review activities:
Army - AR, MI
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - AT, AV, ME
Navy - AS, CG, MC, OS, SH
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 59GP-0051-11)

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